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Leibniz-Institut für Analytische Wissenschaften — ISAS — e.V.
Arbeitsgruppe Mehrdimensionale OMICS-Datenanalyse
Dortmund, Germany
Reference: 398_2026

Re: Doktorand:in (m/w/d) — Mehrdimensionale OMICS-Datenanalyse

Dear Hiring Committee,

I am applying for the doctoral position in the Multi-Dimensional OMICS Data Analysis research group. The combination of modalities in this position — phenotypic, microscopic, PET/MRI, MALDI MSI, and LC-MS/MS data — is precisely what I have built an integrated computational framework to handle. The relevant tools are publicly deployed and available without installation:

- Genomics and sequence analysis: vivid-symbolism.vercel.app/search
- Mass spectrometry (MALDI MSI / LC-MS) experiment design and simulation: lavoisier.vercel.app/experiment
- Microscopy image analysis (PNG / JPG / TIFF, GPU-accelerated): hieronymus-omega.vercel.app/observe/microscopy

Multi-dimensional OMICS integration. The Gospel framework processes whole-genome variant data at 10^6 variants per second, integrates RNA-seq quantification, and applies Bayesian network inference and Gaussian mixture models for multi-omics statistical modelling. It has been validated against 1000 Genomes Phase 3, gnomAD v3.1.2, and UK Biobank, and covers the full pipeline from variant calling through pathway analysis (KEGG, Reactome) to functional annotation. The Lavoisier framework provides the MALDI MSI and LC-MS/MS layer: a dual pipeline combining numerical MS1/MS2 analysis with a CNN-based visual pipeline (PyTorch) that converts spectra into temporal image sequences, achieving 98.9% feature extraction accuracy against the NIST reference library and supporting memory-mapped I/O for datasets exceeding 100 GB. For the microscopy modality, the Helicopter platform fuses fluorescence, brightfield, phase-contrast, and spectral channels through sequential constraint exclusion across twelve measurement modalities, reducing structural ambiguity from $\sim 10^{60}$ configurations to a unique assignment. Together these three systems address five of the six data types listed in this position within a shared S-entropy coordinate framework that makes cross-modal registration tractable.

AI-based personalized risk assessment. This is where the theoretical work I have developed is most directly applicable. The Syndrome framework computes personalised disease state as a vector $\mathbf{D} \in [0, 1]^8$ across eight cellular oscillator classes (protein, enzyme,

channel, membrane, ATP, genetic expression, calcium, circadian), each with a coherence index $\eta_i \in [0, 1]$. The framework locates the patient's current state within a partition geometry on the $[0, 1]^8$ hypercube, computes an $\mathcal{O}(N \log N)$ trajectory to the healthy attractor, and selects the minimum-side-effect therapeutic intervention via sparse ℓ_1 linear programming. This is a personalized risk assessment model in the sense the position describes: patient-specific state, validated biomarker readouts as coherence indices, and a computed trajectory rather than a population-level risk score.

A companion paper, *On Disease Epistemology in Blind Receiver* (affiliated with the AIME Registry and TUM School of Life Sciences Weihenstephan), proves that loop-holonomy self-consistency is the unique viable disease diagnostic criterion, achieving $\text{AUC} = 1.00$ versus 0.896 for template comparison with 25/25 validation experiments passing, and providing a sparse LP for therapeutic target selection.

Large language models and AI tooling. The position specifically lists LLM experience. Beyond using LLM APIs, I have formalised a training principle for domain-specific models: the Purpose framework proves that training exclusively at the hardest single-objective regime (the pure-intent regime) yields competence across all constrained sub-regimes by feasibility inclusion rather than generalisation — with $(N+1) \times$ sample savings and an exponential $2^{N-1} \times$ saving for binary-nested constraint cascades. The Four-Sided-Triangle system implements an 8-stage RAG pipeline with adaptive hybrid LLM and mathematical solver selection and a Rust backend providing 10–50 $\times$ speedup for compute-intensive stages. For the OMICS context this means: not deploying a general-purpose model on specialised data, but training or selecting a model whose capability envelope was defined by the hardest instances of the target domain.

Background. I hold an MSc in Computational Biology (Universität Tübingen) and an MSc in Pharmaceutical Biotechnology (HAW Hamburg), and I completed doctoral research in Computational Lipidomics at TUM / Bayerisches Forschungsinstitut, covering mass spectrometry pipelines, federated learning for distributed clinical analysis, and multi-omics modelling. I am legally resident in Germany with full work authorisation and am available immediately. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye